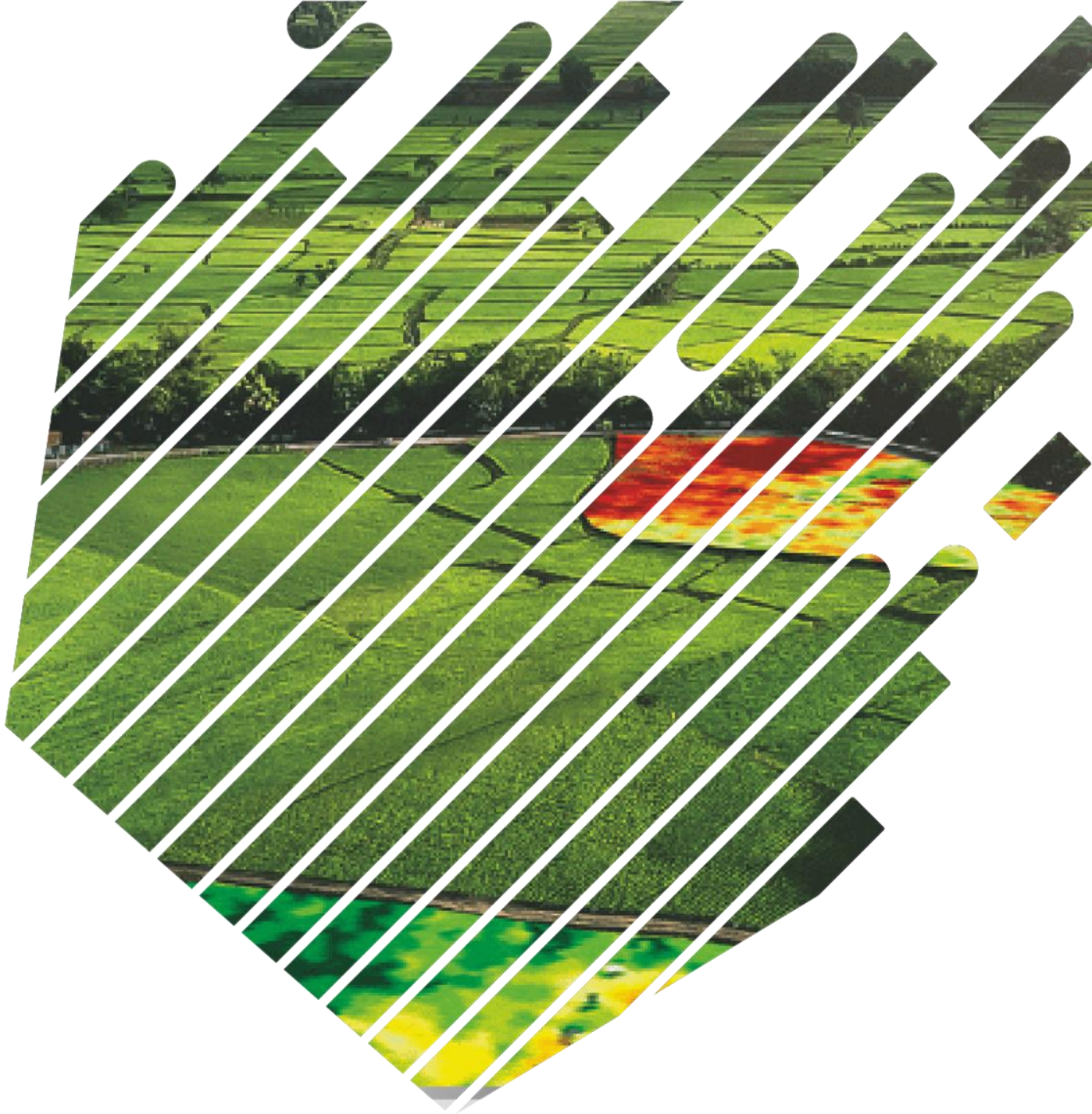
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HARNESSING SATELLITE, SURVEY, AND EVENT DATA IN GEOSPATIAL IMPACT EVALUATIONS: A CASE STUDY

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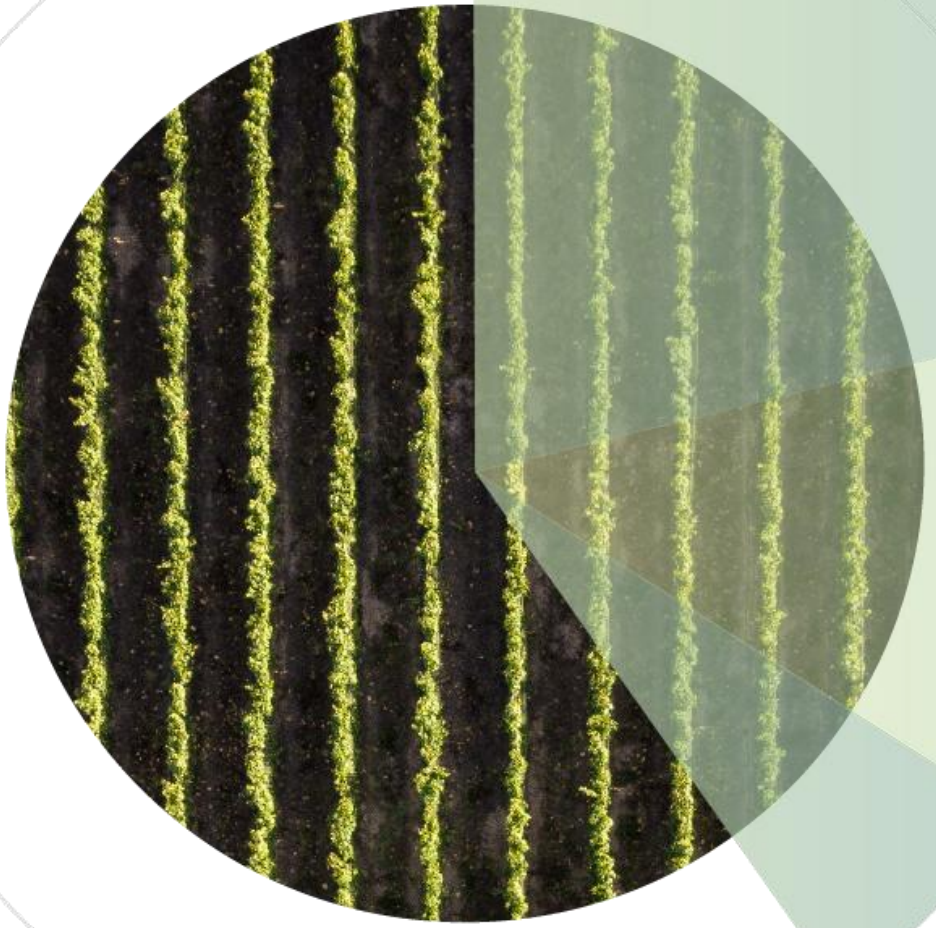
LEARNING OBJECTIVES

- Identify outcomes from a theory of change and determine appropriate data sources for outcome measures
- Convert hard-copy documentation of treatment locations to shapefiles
- Utilize survey, event, and remotely sensed data
- Estimate treatment effects utilizing event study with two-way fixed effects
- Estimate treatment effects utilizing difference-in-differences with two-way fixed effects

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INTRODUCTION

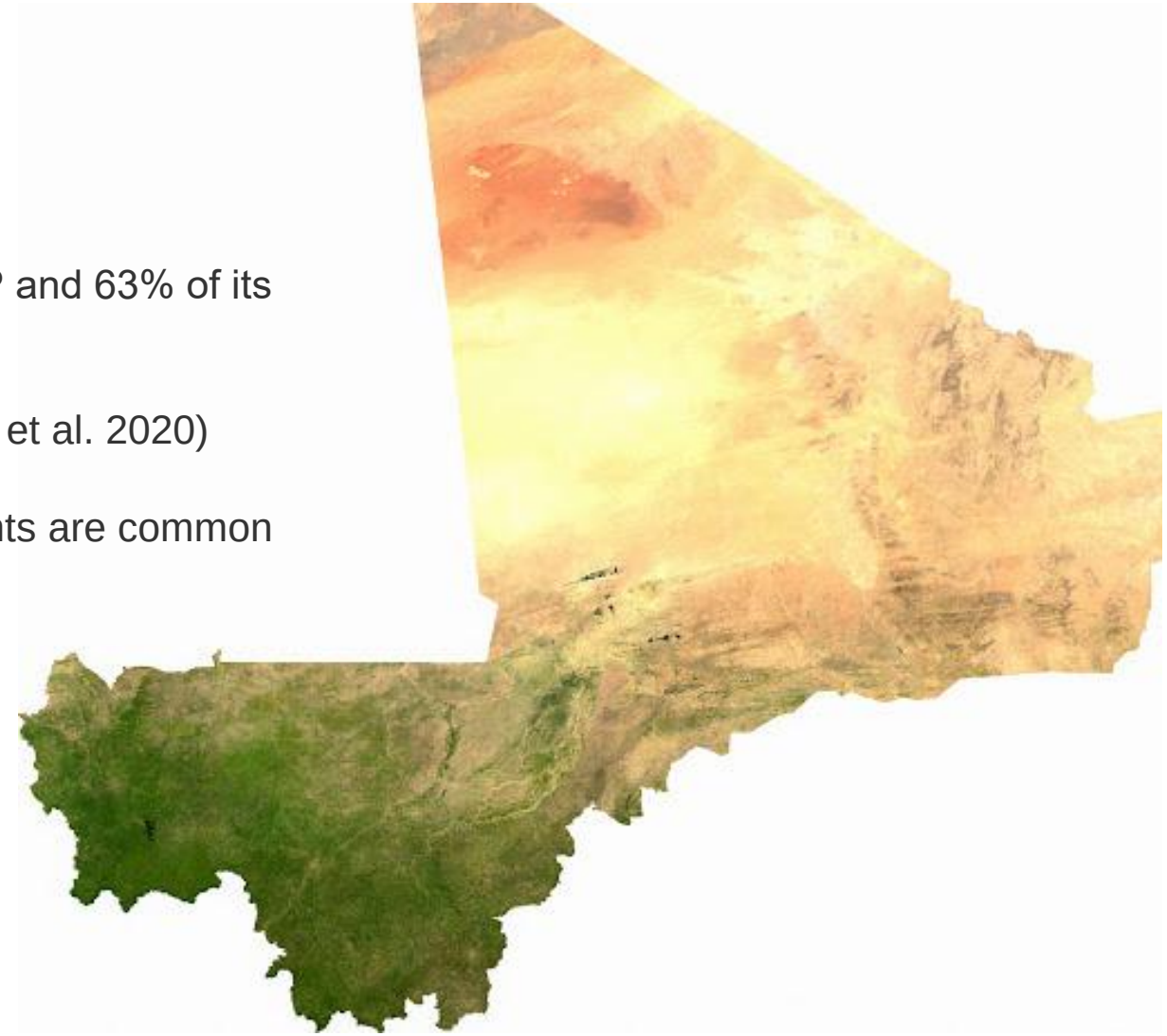
- Agricultural production in the Sahel and sub-Saharan Africa is primarily rainfed
- Rainfed agriculture is particularly vulnerable to climate shocks
- Smallholder farms, crucial to community livelihoods and health, are especially at risk
- Improving access to and use of non-rainfed irrigation stands to improve community resilience
- This case study is a geospatial impact evaluation (GIE) evaluating the impact of irrigation interventions in Mali
- This case study provides experience utilizing remotely sensed, survey, and event data



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STUDY AREA

- Agricultural production accounts for 39% of Mali's GDP and 63% of its labor force employment (Hegazi et al. 2021)
- 95% of Mali's agricultural production is rainfed (Nkonya et al. 2020)
- Rainfall is highly variable in the wet season, and droughts are common (McSweeney et al. 2010)
- Climate change will decrease precipitation, increase drought frequency, and increase temperatures (PIK 2020)
- Mali's Inner Niger Delta Region is an inland floodplain with seasonal flooding



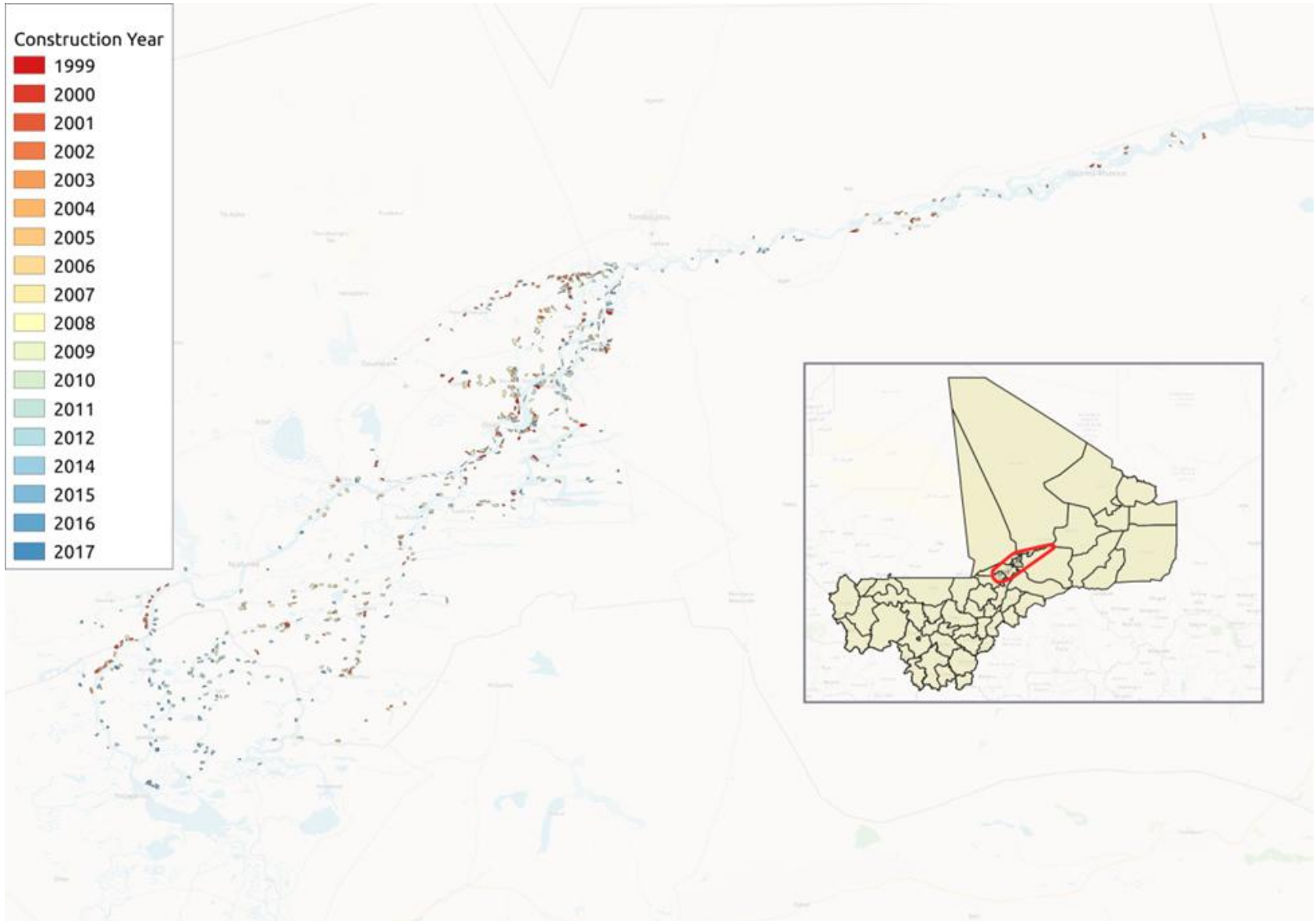
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INTERVENTION

- Nearly 1,000 distinct irrigation intervention sites in northern Mali funded by KfW and GIZ
- Rolled out over past two decades
- Two types of intervention:
  - Pump-based irrigation from Niger River and tributaries to small perimeters
  - Valorization of floodplains (“mares”)
- Expected outcomes:
  - Increased agricultural productivity
  - Increased nutrition and health
  - Reduced conflict



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Construction Year
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2014
2015
2016
2017

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CAUSAL IDENTIFICATION

- Staggered rollout allows us to use difference-in-differences without pure controls

- Utilize two-way fixed effects to absorb:
  - Time-invariant unobservables specific to each spatial unit
  - Shared temporal shocks

- Utilize Chaisemartin and d'Haultfoeuille (2020) estimator

- When considering agricultural productivity, include precipitation and temperature controls

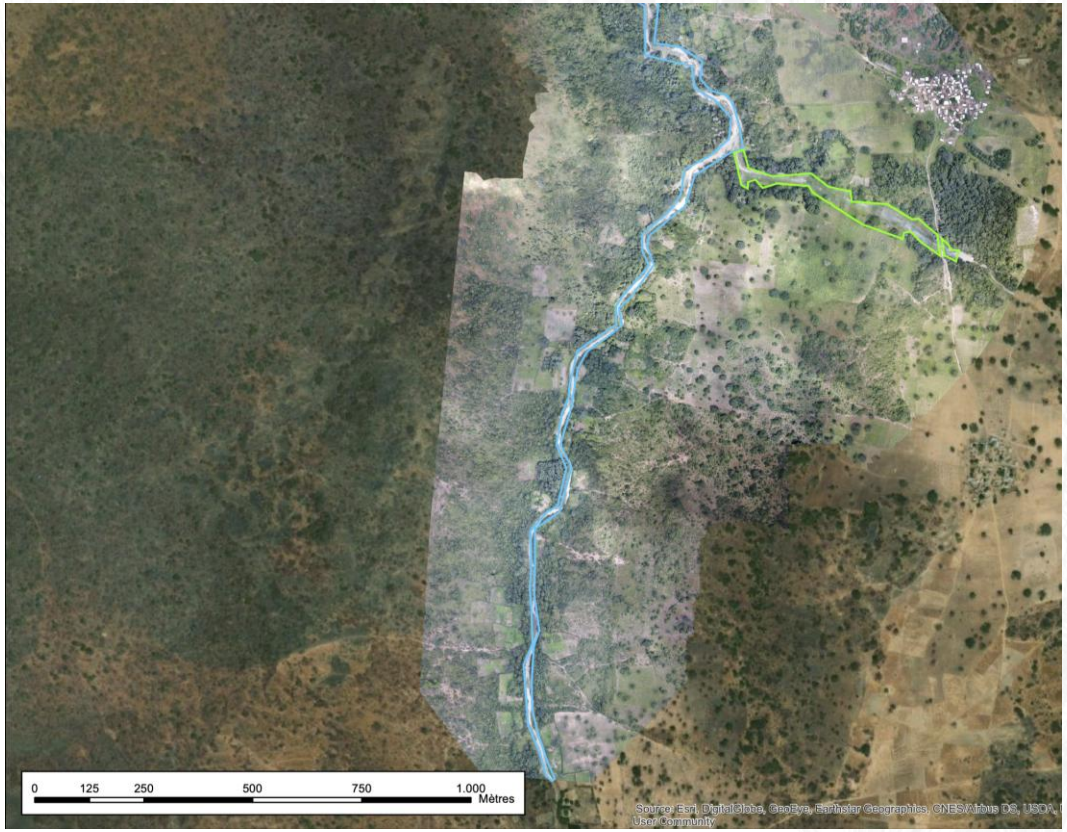
- When considering health and conflict, consider heterogeneous treatment effects by distance band

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OUTCOME DATA

- Agricultural productivity: Remotely sensed NDVI and NDWI from Landsat 5, 7, and 8
- Health: Biometric survey height-for-age and weight-for-age for children under 5 from Mali Demographic and Health Survey (DHS) waves 1996, 2001, 2006, 2013, and 2018
- Conflict: Event data from Armed Conflict Location and Event Data (ACLED)

EXERCISE 1: CONVERT PDF DOCUMENTATION OF TREATMENT LOCATIONS TO SHAPEFILES



- Read and interpret project documentation on intervention locations
- Georeference image in QGIS
- Identify ground control points
- Draw vector lines
- Save as shapefile

*Artificial polygons to protect confidentiality of site locations

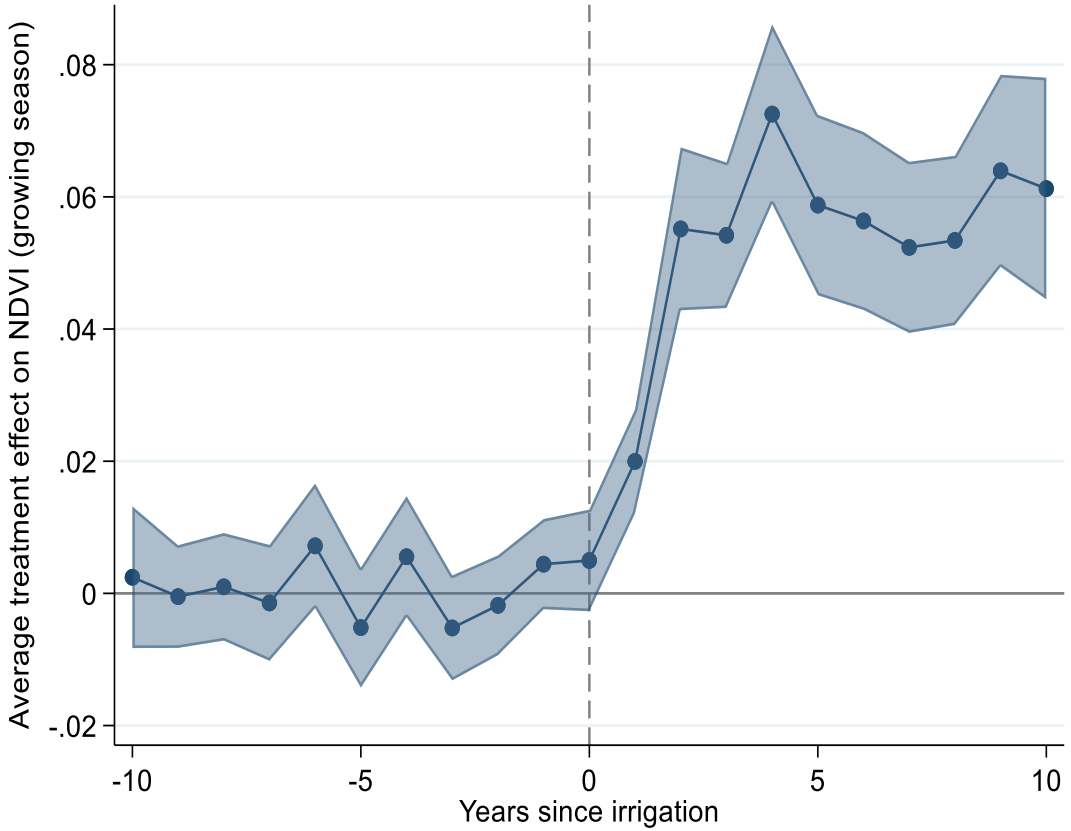
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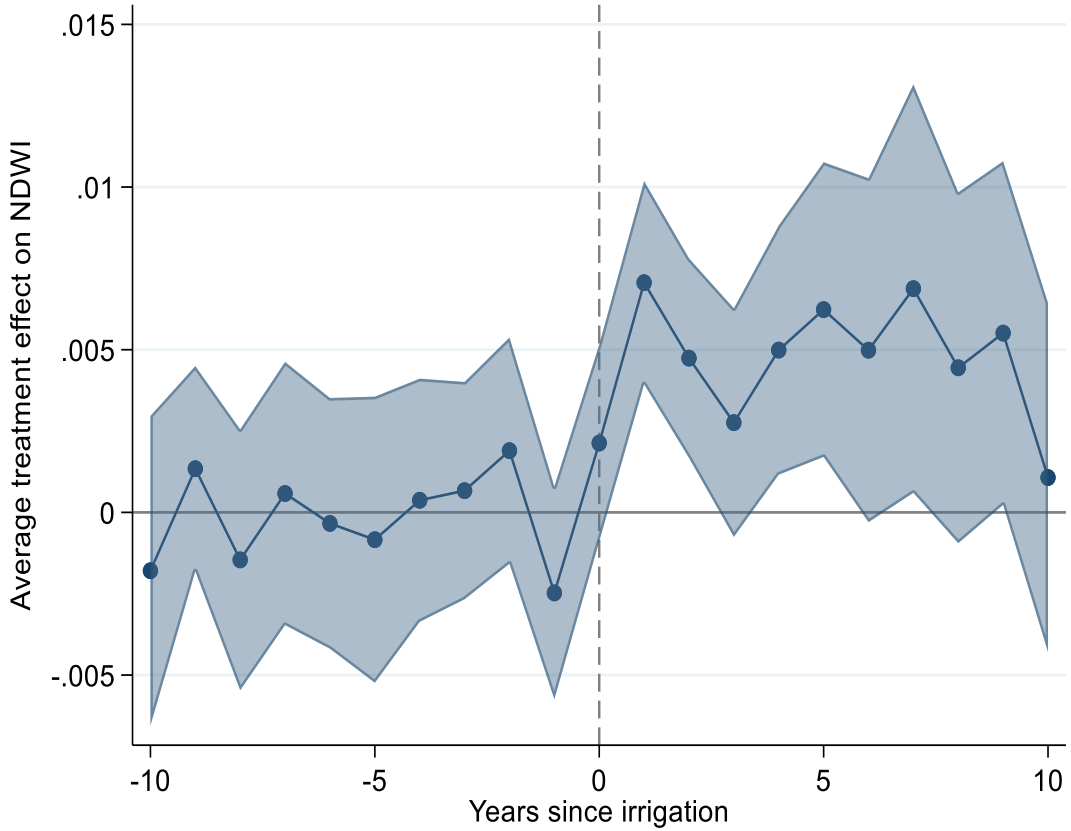
EXERCISE 1: CONVERT PDF DOCUMENTATION OF TREATMENT LOCATIONS TO SHAPEFILES



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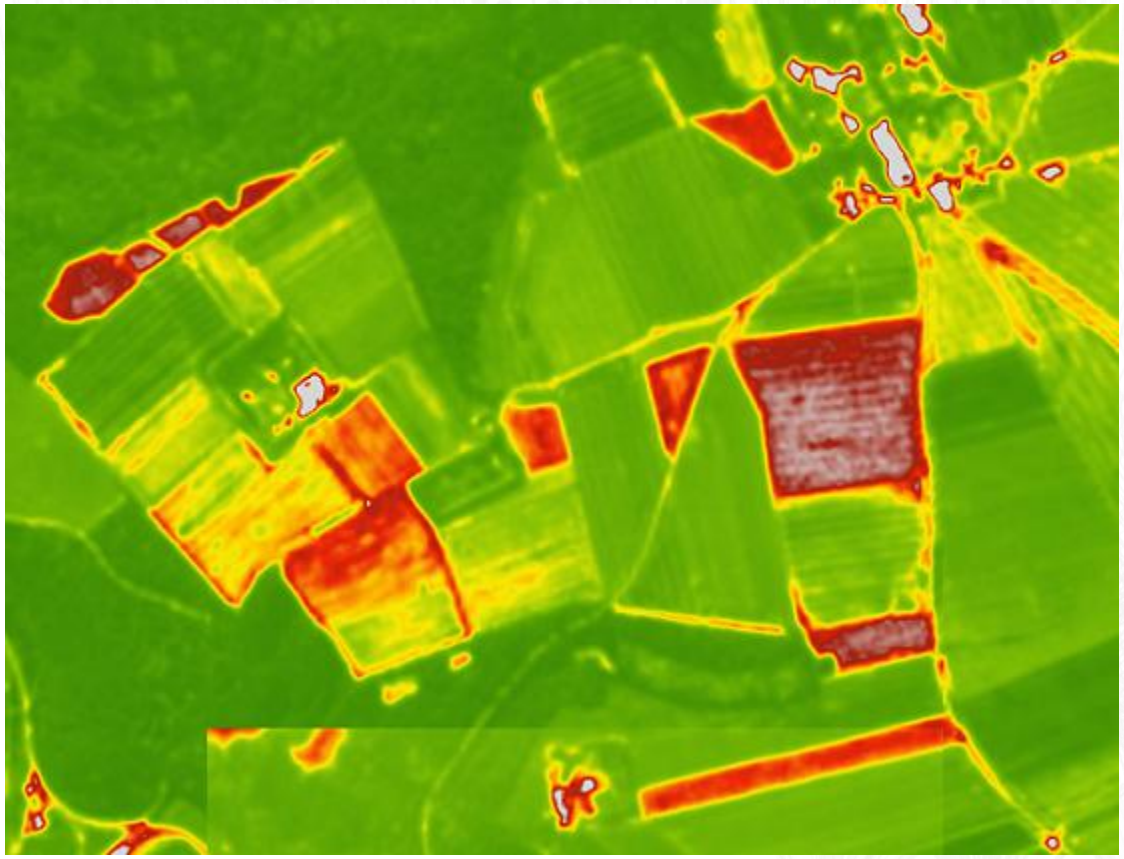
EXERCISE 2: GENERATE NDVI AND NDWI THEN ESTIMATE TREATMENT EFFECTS





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EXERCISE 2: GENERATE NDVI AND NDWI THEN ESTIMATE TREATMENT EFFECTS



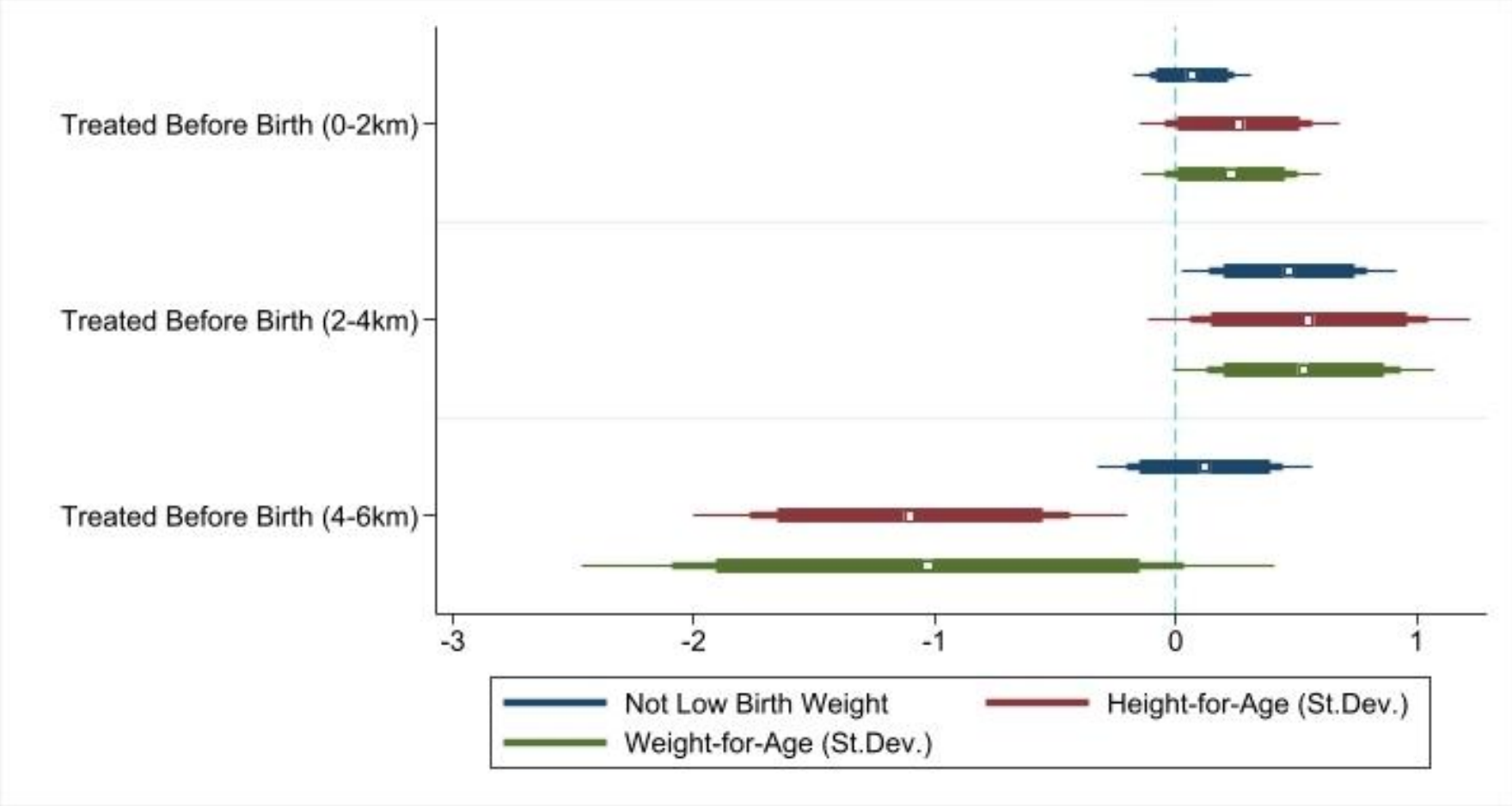
- Load polygons in Google Earth Engine (GEE)
- Write GEE script to generate NDVI and NDWI from Landsat
- Export NDVI and NDWI in csv format
- Estimate treatment effects using an event study design with two-way fixed effects in Stata
- Estimate treatment effects using Chaisemartin and d'Haultfoeuille (2020) estimator in Stata

*Artificial polygons to protect confidentiality of site locations

*Anonymized dataset with randomly generated site IDs and randomly perturbed NDVI and NDWI

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EXERCISE 3: UTILIZE SURVEY DATA TO ESTIMATE TREATMENT EFFECTS ON HEALTH



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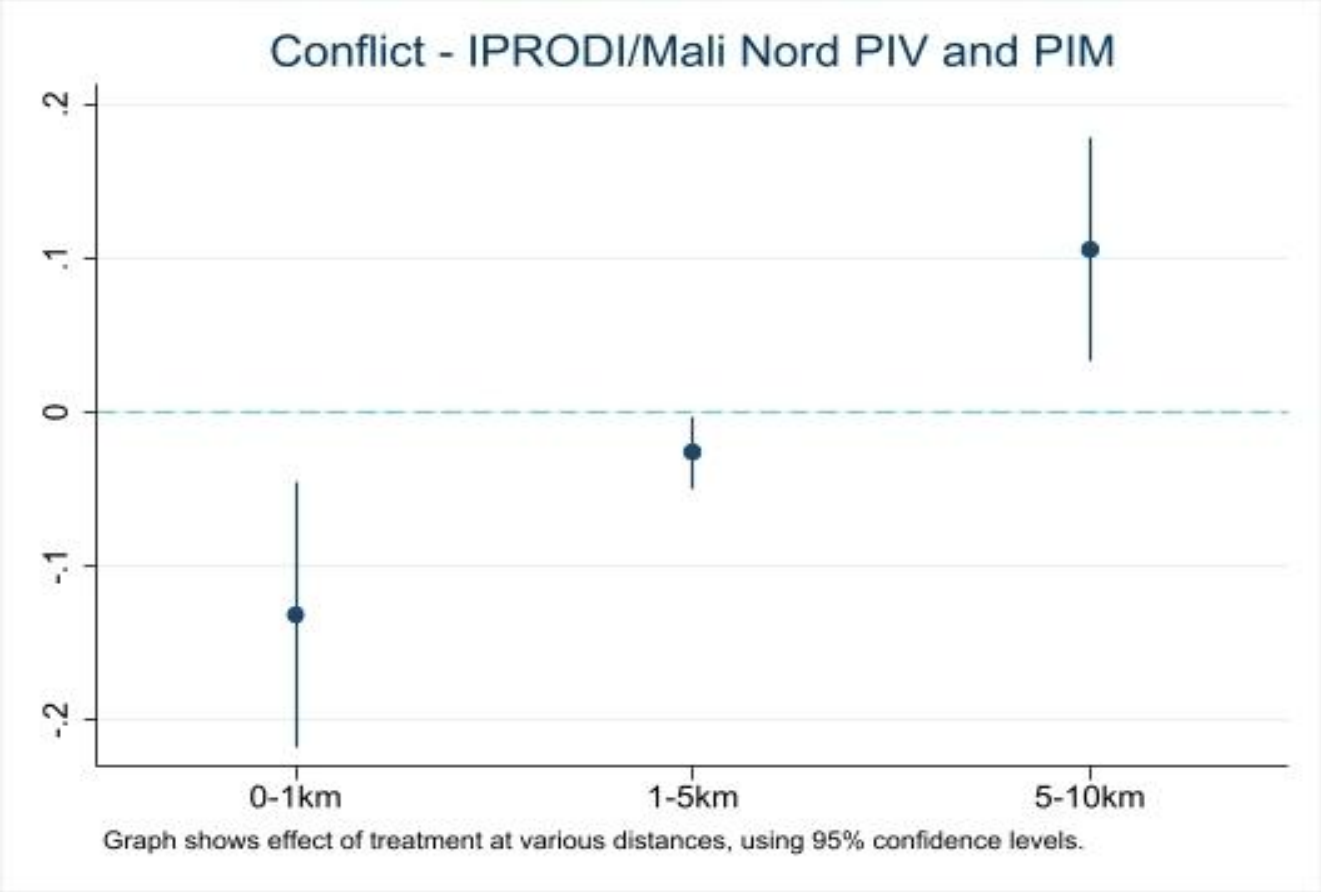


- Utilize Stata to create treatment by distance band interactions
- Estimate effect of treatment before birth on child stunting and wasting using difference-in-differences with two-way fixed effects
- Compare to effect of treatment before survey on child stunting and wasting
- Interpret regression results

*Anonymized dataset with randomly generated site IDs and randomly perturbed health data

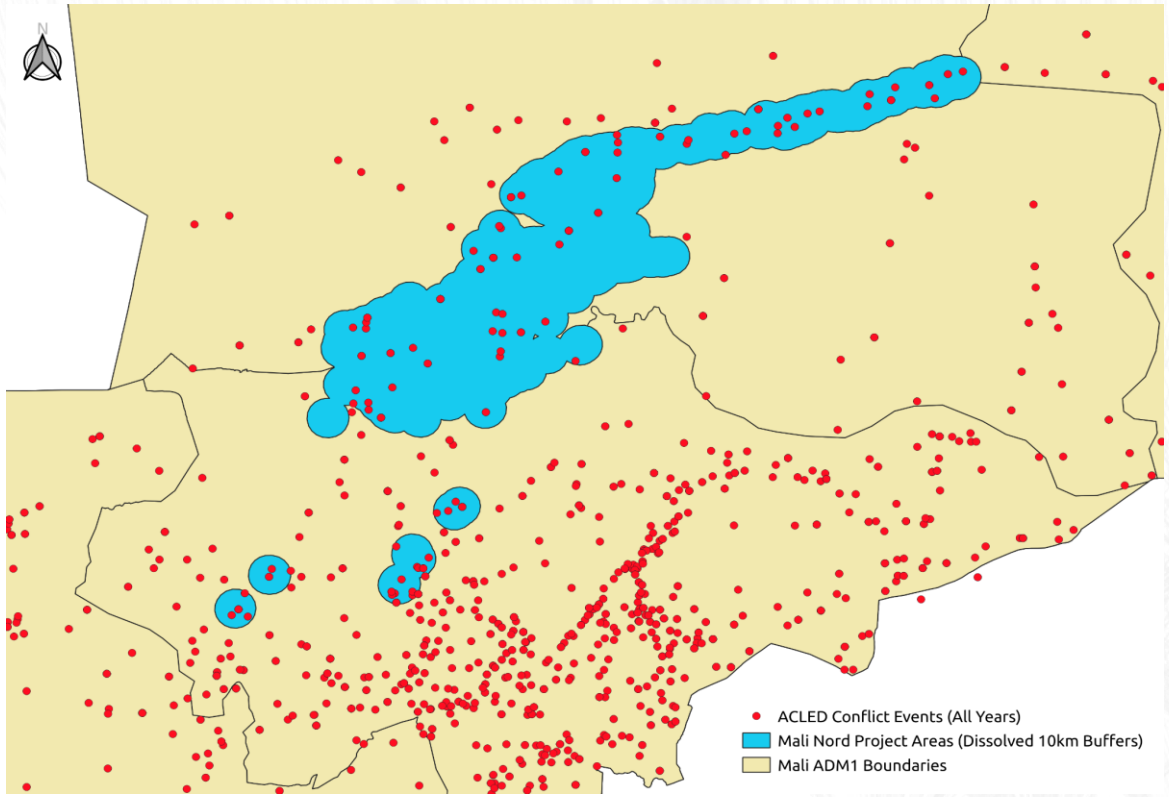
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EXERCISE 4: GENERATE CONFLICT COUNTS USING BUFFER APPROACH AND ESTIMATE TREATMENT EFFECTS



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EXERCISE 4: GENERATE CONFLICT COUNTS USING BUFFER APPROACH AND ESTIMATE TREATMENT EFFECTS



- Write Python script to load treatment polygons and conflict event data
- Utilize buffer approach to generate nearby conflict counts
- Use Stata to estimate treatment effect on conflict events using difference-in-differences with two-way fixed effects
- Interpret heterogeneous results by distance

*Artificial polygons to protect confidentiality of site locations

*Anonymized dataset with randomly generated site IDs and randomly perturbed conflict data (maintaining lower bound of 0)